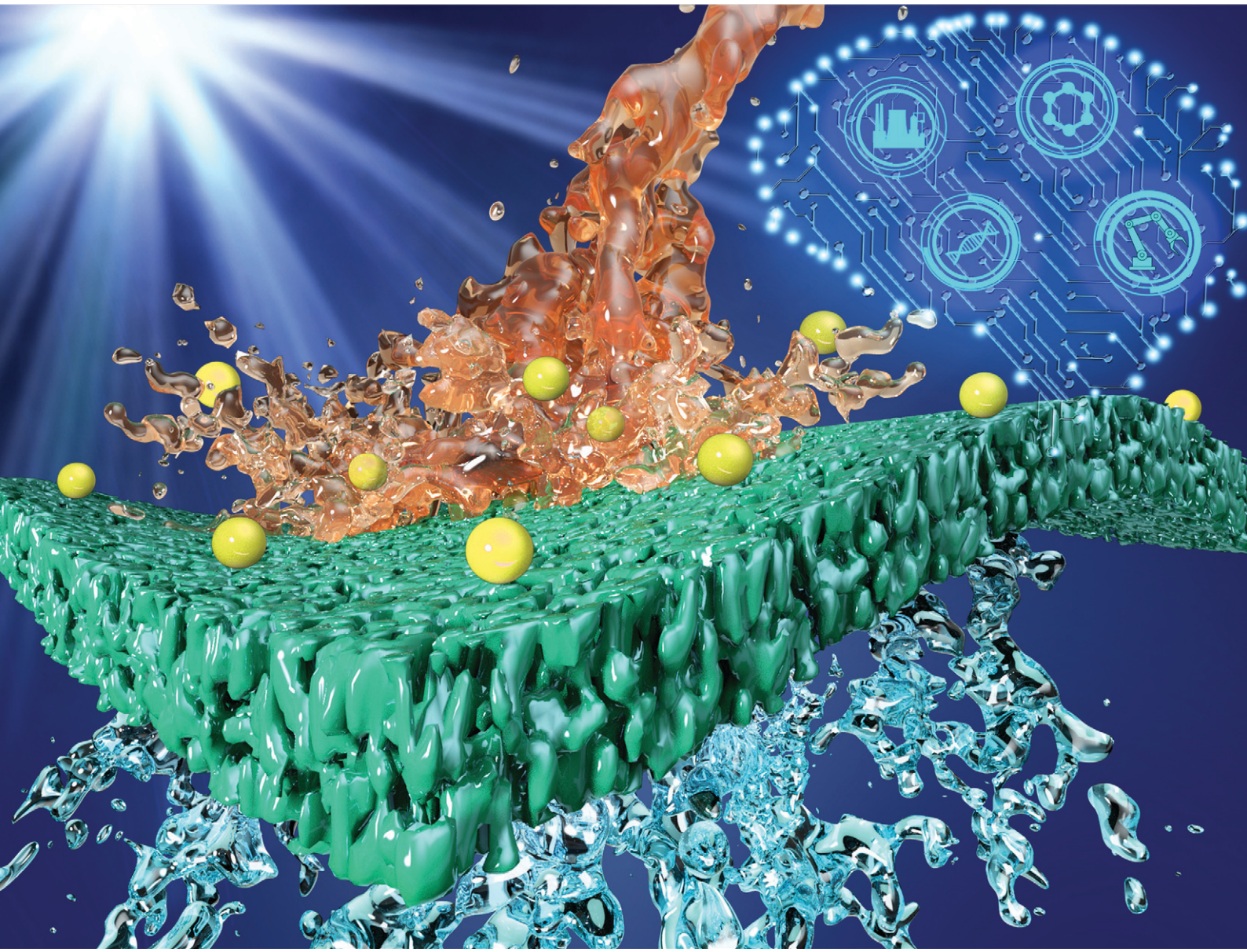
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Recent development in machine learning of polymer membranes for liquid separation

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Emerged as a transformative technology, machine learning (ML) has demonstrated unprecedented success in the design and discovery of new materials. Over the past few years, we have witnessed the increasing applications of ML in the field of membranes. In this review, we present recent development in ML studies of polymer membranes for liquid separation, including water treatment at both process and membrane levels, pervaporation based water–organic and organic–organic separation, and solvent recovery. Along with these latest advances, we suggest directions for future ML exploration: (1) Membrane Separation Genome, (2) Membrane and process representations, (3) Deep learning algorithms, (4) ML enhanced high-throughput approach, and (5) Digital Membrane Separation. These directions provide new opportunities and challenges in this rapidly evolving field to accelerate membrane development for high-performance liquid separation.

Design, System, Application

With excellent scalability, reproducibility and solution-processability, polymer membranes have been the dominant materials for large-scale gas and liquid separation. Given their amorphous nature, their atomistic structures are poorly defined and difficult to investigate experimentally. These challenges have motivated the development of computational capabilities and algorithms to rationally design high-performing polymer membranes and understand their separation mechanisms. Over the past few years, machine learning (ML) has demonstrated increasing potential in the modelling of polymer membranes for liquid separation. This review summarizes the recent applications of ML in this vibrant field to characterize structures of polymer membranes, unravel fundamental insights, design new membranes, and predict their performance for liquid separation. Particularly, ML studies for water treatment, pervaporation and solvent recovery are presented and discussed. While the progress and achievements of ML efforts have been promising to date, new opportunities and challenges exist. Future ML exploration is expected to facilitate the holistic rational design of membrane materials, separation modules and plantwide processes, thus accelerating the commercialization of new polymer membranes toward high-performance separation.

1. Introduction

Separation is a basic and crucial unit operation in the chemical and pharmaceutical industries.[1] By leveraging on the difference in physicochemical properties of constituents present in a mixture, separation can be achieved. In many chemical processes, 40–70% of capital and operating costs are typically incurred by separation.[2] Moreover, energy usage for separation may constitute as high as 10–15% of the world's energy consumption.[3,4] In the U.S. alone, separation accounts for half of industrial energy usage, in which 80% is consumed by energy-intensive thermal separation technologies such as distillation.[5] In contrast to traditional separation processes, membrane technology offers several competitive advantages. Without the requirement of phase transition, membrane separation consumes significantly less energy, increases separation efficiency, reduces degradation of thermolabile molecules, and minimizes capital cost and chemical wastage.[6] Different types of organic, inorganic and hybrid materials have been developed into membranes to enable highly efficient separation.[7] Meanwhile, process intensification is also being accelerated by membrane technology.[8] Due to its modular and scalable nature, membrane separation can be easily integrated into existing unit operations to simplify processes.

In view of their low cost, scalability and reproducibility, polymers are the currently dominant materials being fabricated into membranes. Their excellent solution-processability also facilitates membrane formation through different methods. More importantly, through judicious selection and combination of a wide range of monomers, polymer membranes can be uniquely synthesized with tailored separation performance. Many of them have been

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and continue to be utilized for large-scale gas and liquid separation.[9–11] Furthermore, polymer composite membranes have been fabricated by incorporating functional nanofillers into polymer matrices to discriminate molecularly similar chemicals.[12–14]

Being amorphous in nature, the atomistic structures of polymer membranes are poorly defined and challenging to investigate experimentally. With advanced development of computational capabilities and algorithms, computational methods have been increasingly adopted to examine polymer membranes. In the last 1–2 decades, molecular simulation has demonstrated tremendous success in characterizing polymer membranes, microscopically understanding separation mechanisms, and rationally designing new membranes, particularly for gas separation such as CO_2 capture.[15,16] In terms of liquid separation through polymer membranes, molecular simulation is well established for water treatment.[17,18] For non-aqueous separation, however, only recently have a few molecular simulation studies been reported.[19]

In the past several years, machine learning (ML) has brought unprecedented success across many scientific and technological fields.[20,21] It has garnered widespread interest as a promising computational tool for various disciplines of chemical and physical sciences.[22] Similarly, the field of polymer membranes is poised to benefit from the immense potential of ML,[23] particularly with the establishment of polymer databases and informatics platforms.[24] Recently, ML models have been developed on polymer membranes for both gas and liquid separation.[25] This review aims to demonstrate the prowess of ML for liquid separation. As practical liquid separation through polymer membranes involves various intricately connected membrane properties and processes, the description of these complex phenomena across multiple length and time scales has not been straightforward. Therefore, the unique relevance and potential of ML in describing and modelling polymer membranes for liquid separation are discussed here. We particularly showcase how ML methods are used to characterize membrane structures, unravel fundamental insights, predict membrane performance, and design new membranes. In section 2, two main ML methods (supervised and unsupervised learning) are briefly described. Recent ML studies for water treatment, water–organic and organic–organic separation, and solvent recovery are critically summarized in sections 3–5. For each liquid separation, the main ML algorithms used are introduced, and the current progress and challenges are discussed. Finally, directions for future ML exploration are suggested in section 6, which also highlights new opportunities and challenges in this vibrant field.

2. ML methods

ML is a critical data-driven approach to design functional materials including polymers.[23,26] By utilizing various physically meaningful descriptors of different natures and scales, ML has enabled the prediction of a range of polymer properties such as the glass transition temperature, refractive index and dielectric constant.[27–31] Similarly, polymer membranes have also recently been investigated through ML for gas separation.[32–35] Broadly, there exist two ML methods, namely supervised and unsupervised learning.[36] Fig. 1 provides an overview of the two ML methods and related algorithms. For the detailed description and implementation of each algorithm, we direct interested readers to excellent textbooks on ML.[36,37]

In supervised learning, a training dataset comprises input and corresponding output data.[37] Through the labeled training dataset, supervised learning establishes a mapping



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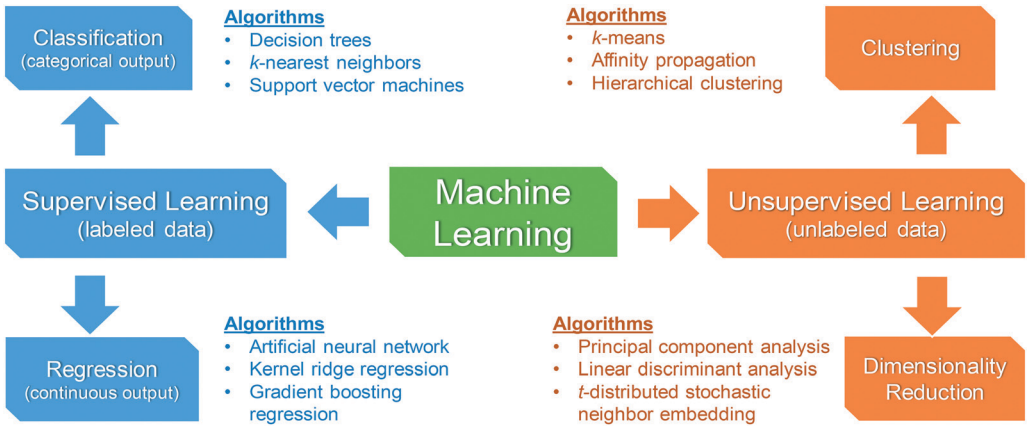


Fig. 1 ML methods and related algorithms.

function from input to output *via* a classification or regression algorithm. A classification algorithm predicts discrete and categorical output, while a regression algorithm estimates continuous and numerical output. Based on the attributes in input data, classification categorizes objects into one of several defined classes. By optimizing defined loss functions (*e.g.*, cross-entropy and hinge loss), accurate classification models can be obtained and their capabilities are evaluated through various predictive metrics such as the receiver operating characteristic and the area under the curve. Common classification algorithms include support vector machines (SVMs) and decision trees (DTs).[37] For regression, a predefined mathematical function is trained with optimal model coefficients and then used to make new predictions. To train these models, an objective function that employs the least squares error between actual and predicted outputs is minimized. Regularization is also introduced in the objective function to prevent overfitting and control model complexity. Linear models with linear functions of input data tend to be simpler and more interpretable, *e.g.*, multiple linear regression and least absolute shrinkage and selection operator regression (LASSO).[38] On the other hand, nonlinear regression provides more accurate predictive models based on functions of neurons, kernels and trees.[36] For all supervised learning, the resampling *k*-fold cross-validation is performed to select the best ML models with optimal hyperparameters.

In contrast, unsupervised learning has no labeled dataset; only input data are present in a training dataset from which patterns are to be inferred.[37] Without the availability of corresponding output data, input data alone are difficult to visualize and understand often due to their high dimensionality nature. As such, unsupervised learning can automatically reveal inherent structure from input data without much human guidance. Common unsupervised learning methods include dimensionality reduction, clustering and pattern recognition, which facilitate the representation of a large number of descriptors into a low-dimensional space. For example, the dimensionality reduction algorithm identifies the dominant orthogonal directions of data variance and reduces data to the first few principal components. On the other hand, clustering analysis allows the segmentation of different objects into subsets based on defined similarity measures.

By implementing the aforementioned ML algorithms, the design and optimization of polymer membranes for liquid separation at the membrane and process levels can be achieved. At the membrane level, ML quantifies the influence of membrane fabrication parameters on membrane morphologies and separation performance. The complex interplay of interactions among membranes, solvents and solutes that affect liquid separation can also be analyzed by ML with accurate prediction at minimal human guidance. For example, fabrication conditions, structural and physicochemical properties of membranes, solvents and solutes can be employed as input variables of artificial neural networks (ANNs) to predict liquid separation performance with little *a priori* knowledge. At the process level, search algorithms such as genetic algorithms (GAs) may also be applied in addition to ML algorithms. Based on trained ML models, the probabilistic search algorithms can efficiently identify key operating parameters and globally optimize the automation and management of unit operations in industrial process plants. However, we should also note that ML models require careful hyperparameter tuning and are less physically interpretable compared with classical theories and molecular simulations of membrane separation in polymer membranes, which invoke established physicochemical laws. Furthermore, to be accurate and robust, data-driven models typically require large datasets. In the current ML of polymer membranes for liquid separation, datasets typically span from a few hundreds to tens of thousands in size. The number of samples is dependent on the nature and source of datasets including the diversity of polymer membranes, the realistic range of fabrication and operating conditions of membranes, and the measurable properties of membranes, solvents and solutes. In addition, the dataset size is also influenced by the ease of obtaining membrane performance such as membrane swelling, solvent permeability and solute rejection.

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3. Water treatment

Clean water scarcity is currently one of the major societal challenges. It is further exacerbated by the global increase in human population, drinking water contamination, accelerated industrialization and climate change.[39] Polymer membranes[40,41] and mixed-matrix membranes[42] (MMMs) are the dominant materials used for water treatment. While numerous molecular simulations have been carried out on these membranes to fundamentally investigate their separation performance,[43] ML studies have been recently reported at both process and membrane levels.

At the process level, Hilal and coworkers discussed the recent progress and application of ML to process automation for water treatment. ANNs and GAs were highlighted as the two main techniques employed in the field for regression and global optimization, respectively. They summarized ML studies for water treatment, including the modeling and optimization of ion and pollutant removal and the cost and efficiency of desalination and wastewater plants, as well as membrane property–performance relationships. Unlike conventional phenomenological modeling, these ML studies offered the formulation of nonlinear relationships between operating conditions and separation performance.[44] On the advanced control of membrane fouling, Bagheri *et al.* summarized several ML approaches for water and wastewater treatment. They first identified the essential parameters (*e.g.*, operating conditions, membrane and biomass characteristics) contributing to fouling in membrane bioreactors as inputs for ML. Then, individual models including ANNs, fuzzy logic, genetic programming and model trees were discussed. Upon comparison, they concluded that hybrid models comprising individual models and optimization methods (GA and particle swarm optimization) were more favored for their rapid optimization of model parameters. The incorporation of clustering analysis, image recognition and feature selection to improve the monitoring of membrane fouling was also reviewed.[45] Yu and coworkers discussed various applications of ML tools in drinking water treatment plants. They first summarized the common ML techniques and search algorithms employed to investigate water quality, coagulation/flocculation, disinfection and membrane filtration. Not only was the prediction performance between ML and traditional models critically compared, but the challenges in the application of ML methods in drinking water were also highlighted. Furthermore, they proposed novel detection methods to analyze water quality and the need to develop a macro model for drinking water treatment plants *via* ML.[46]

Relatively, ML studies at the membrane level for water treatment are limited. By employing ANNs and GAs, Rahimpour and coworkers examined and optimized the fabrication of ultrafiltration membranes. They prepared polyethersulfone and polysulfone membranes by immersion precipitation using different types of solvents, nonsolvents and additives. All fabrication parameters and polymer and additive concentrations were first dimensionally reduced before using as inputs for an ANN model. Then, an optimal two hidden layer ANN was identified to accurately predict flux and rejection. Subsequently, a GA was employed to guide the experimental preparation of the optimal membrane with the highest flux and rejection.[47] From the published data of thin film nanocomposites (TFNs) for reverse osmosis (RO), Yeo *et al.* also adopted ML to understand and optimize membrane preparation and separation. To predict the water permeability and salt pass rate through these membranes, a gradient boosting tree model was developed based on the properties of composite membranes and nanoparticles (loading, size, pore size and shape). They showed that porous nanoparticles performed better than their nonporous counterparts and identified key factors in membrane performance, namely the loading, size and hydrophilicity of nanoparticles. Then, partial dependence plot analysis was utilized to further illustrate the relative importance of various membrane parameters.[48] To predict ion separation performance, Wessling and coworkers developed a hybrid model consisting of ANN and membrane transport models for layer-by-layer polyelectrolyte membranes. From an extensive experimental dataset of four input fabrication parameters (bilayer number, salt concentration, coating flux and time), an ANN model was trained to provide accurate prediction of separation performance. Local single- and global multi-objective optimizations were employed to improve the fabrication conditions of existing membranes and even identify their upper bound of the trade-off between ion rejection and water permeability. Furthermore, an extended Nernst–Planck model for the transport of charged species was also incorporated in the ANN model to fundamentally understand the effect of fabrication parameters on separation.[49]

As the design of the membrane synthesis and water treatment process is typically independent, Wessling and coworkers recently proposed various modeling strategies to simultaneously optimize separation at both process and membrane levels. As shown in Fig. 2, they used ANN and mechanistic process models to propose a simultaneous design framework of the membrane synthesis and process to achieve desirable ion separation performance. With the trained ANN embedded in the process model, the best membrane types and operating conditions were identified through a global nonlinear optimization approach.[50] Such a modeling strategy not only leverages on the available search space and synergy between membranes and processes, but also offers overall optimized membrane performance at a lower cost. Subsequently, they developed a multiscale approach to optimize membrane processes for water treatment. Beginning from a nanoscale level, data simulated from the Nernst–Planck equation for ion transport were trained into an ANN; then the ANN surrogate model was utilized to describe the filtration performance of membrane modules in a process plant. By modeling from the nanoscale

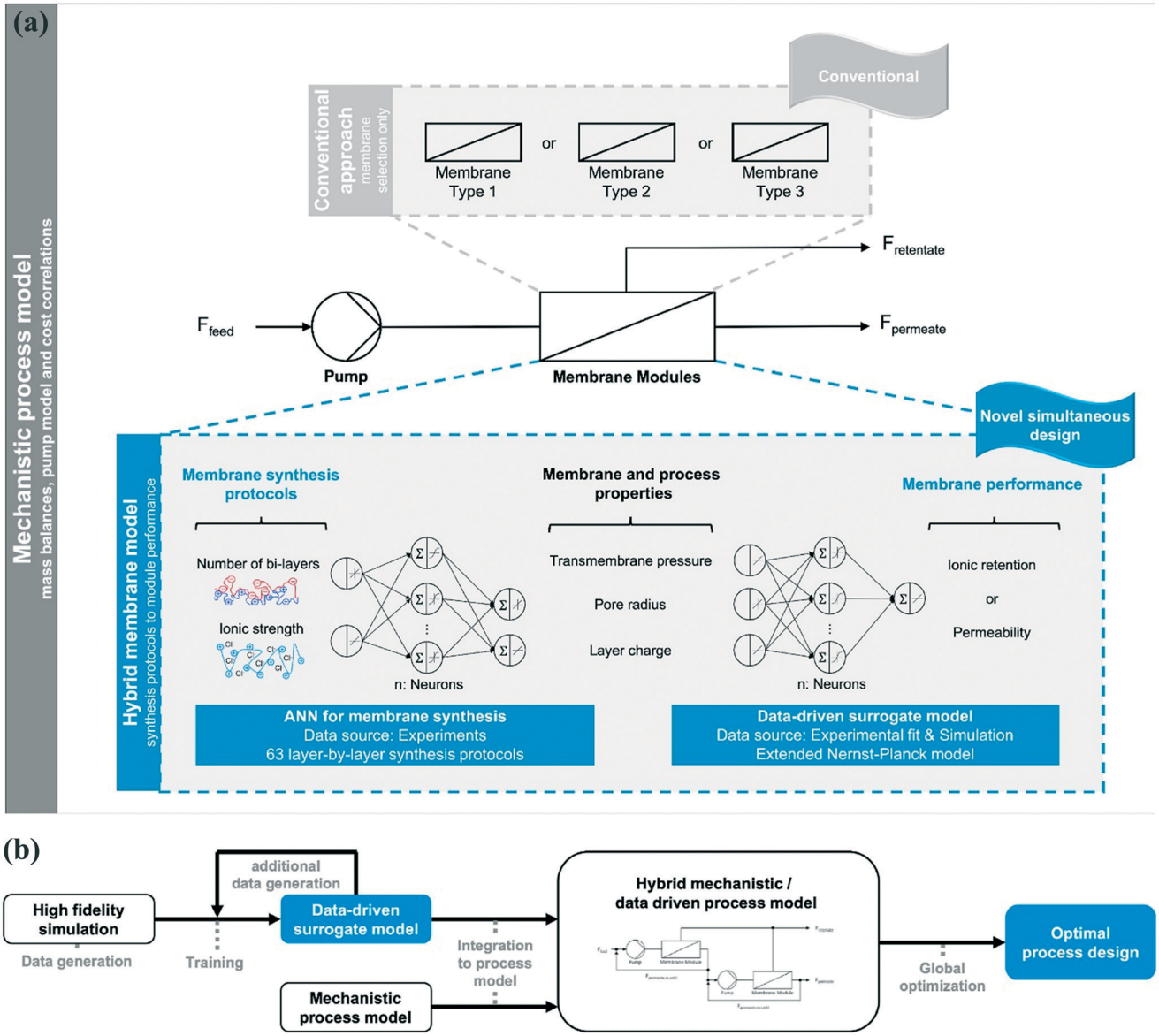


Fig. 2 ML at both process and membrane levels for water treatment. (a) A simultaneous rational design of membrane synthesis and process.[50] (b) A hybrid data-driven/mechanistic model for multiscale optimization.[51] Reproduced with permission: Copyright 2020, Elsevier.

to the process level, they demonstrated the simultaneous optimization of the membrane synthesis protocol and process design to meet multiple separation objectives.[51]

As discussed above, existing ML studies for water treatment are primarily focused on the efficient automation and management of unit operations (*e.g.*, membrane filtration and coagulation) at the process level. Importantly, such a computational approach has demonstrated to be an alternative and accelerated approach over conventional molecular simulations for water treatment. In recent years, ML studies at the membrane level have been limited to the identification of essential preparation conditions to optimize membrane structures with improved separation performance. Meanwhile, ML for multiscale optimization at both membrane and process levels has been also reported. Despite the above mentioned recent progress, several issues should be addressed to further advance ML of polymer membranes for water treatment. Apart from interfacial polymerization, membranes can be also fabricated by other methods (*e.g.*, casting and phase inversion). Therefore, it is essential to understand how different fabrication approaches affect membrane formation, structural transformation and separation performance. Membrane preparation conditions (*e.g.*, temperature, pressure and monomer compositions) have a significant effect on membrane morphologies, which in turn affect separation performance. As these complex relationships are often not straightforward and measurable, ML studies could unravel and quantify the effect of membrane fabrication conditions on membrane morphologies. Membrane fouling is another major issue that

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affects water treatment.[43] For example, foulant adsorption on an irregular membrane surface may lead to many detrimental outcomes such as lower water permeation and concentration polarization. Given the large diversity of organic and inorganic foulants, future ML should explore the detailed foulant–membrane interactions and binding configurations, and their subtle effect on separation performance. To design new membranes for water treatment, molecular simulation has been a well-established computational approach to investigate membrane structures and properties from bottom-up.[43] While such fundamental studies derive invaluable microscopic insights, they are limited to a few specified membranes and incur high computational costs. On the other hand, current ML modeling towards the design of new polymer membranes for water treatment is scarce. In addition to membrane preparation and process optimization, membrane materials play a central role in determining overall separation performance. ML studies based on molecular representation are highly desired to fundamentally establish direct relationships between membrane structures and separation performance, thereby accelerating the rational design of new membranes for high-performance water treatment. Furthermore, as demonstrated by the recent studies in Fig. 2, future ML for water treatment should also aim to adopt multiscale modeling for holistic development and optimization of both membrane synthesis and process design.

4. Water–organic and organic–organic separation

Traditionally, water–organic and organic–organic mixtures are separated by distillation, which however is energy intensive. As an integration of membrane permeation and evaporation, pervaporation (PV) has emerged as a robust separation technology.[52] It enables efficient separation of many challenging aqueous and non-aqueous mixtures, and yet reduces environmental impact, cost and energy demand.[53] For example, the cost to separate an ethanol/water mixture (95/5 wt%) at a capacity of 1000 liters per day was estimated to be $75 000 by PV, which is far more efficient than $140 000 by distillation.[54] These attributes have led to the abundant applications of PV, including dehydration of organics, recovery of organics from aqueous mixtures and organic–organic separation.[55] Polymer membranes and MMMs have been actively fabricated and tested for PV.[10,52] Nevertheless, there are only a handful of simulation studies for PV[19] and ML is scarce.

At present, ML studies for PV have been attempted to identify the most crucial process and membrane preparation conditions that contribute to high PV performance. Farshad *et al.* investigated and optimized PV at the process level for toluene/heptane separation using a commercial polymer composite membrane. Through four nested loops, they identified the best ANN to be a network of one hidden layer with three neurons. Process conditions (temperature, permeate pressure and feed composition) were then used as ANN inputs to predict permeation and selectivity, and excellent agreement was achieved between actual and predicted results. With the trained ANN, the multi-objective GA was finally applied to obtain the Pareto set of optimized process conditions, where temperature and feed composition were the dominant factors.[56] To reveal the effects of process conditions on isopropanol dehydration performance in PVA membranes, Kasiri and coworkers first synthesized PVA membranes crosslinked by glutaraldehyde. They obtained the best ANN with an optimal architecture through the multi-objective genetic optimization algorithm. Four operating conditions (feed composition, operating temperature, permeate pressure and Reynolds number) were investigated as inputs for the ANN to estimate flux and selectivity. Furthermore, the authors demonstrated the extrapolation capability of the optimized ANN and concluded the feed composition as the most influencing factor on PV in PVA membranes.[57]

At the membrane level, Rezakazemi *et al.* developed a robust ANN to predict isopropanol dehydration in PVA–APTEOS/TEOS nanocomposite membranes. Using the Bayesian regularization algorithm, the optimal feed forward backpropagation ANN consisted of two hidden layers with three neurons and a log-sigmoid transfer function. The water concentration, feed temperature and nanoparticle content were considered as the inputs, while the separation index was the output. Particularly, the nanoparticle content was determined to contribute the most to PV. Given the accurately trained ML models, they further applied simulated annealing to identify the optimal PV separation index.[58] Feng and coworkers constructed hybrid models based on an ANN and GA to examine the effects of preparation conditions on the PV performance in PDMS/ceramic composite membranes. They optimized an ANN architecture, containing a tan-sigmoid transfer function and a single hidden layer with 15 and 10 neurons, to predict flux and selectivity, respectively. The inputs were the PDMS concentration, crosslinking agent concentration and dip-coating time, while the outputs were the ethanol flux and ethanol/water selectivity. Connection weight analyses indicated a similar effect of each input on the PV performance. Compared to response surface methodology models, their hybrid ML models were more accurate in predicting the PV performance and informing optimal membrane preparation conditions.[59] Recently, Wang *et al.* developed two ML models to predict and high-throughput screen polymer membranes for PV separation from a curated experimental dataset with 681 data samples consisting of 16 polymers and 6 organic solvents. As shown in Fig. 3, in the first model based on gross descriptors, the water composition in the feed was revealed to be the most influential feature in determining the total flux and separation factor of PV. Furthermore, polymers and solvents were molecularly represented in the second model, which was subsequently applied to screen a polymer database containing ~1 million hypothetical polymers for water/

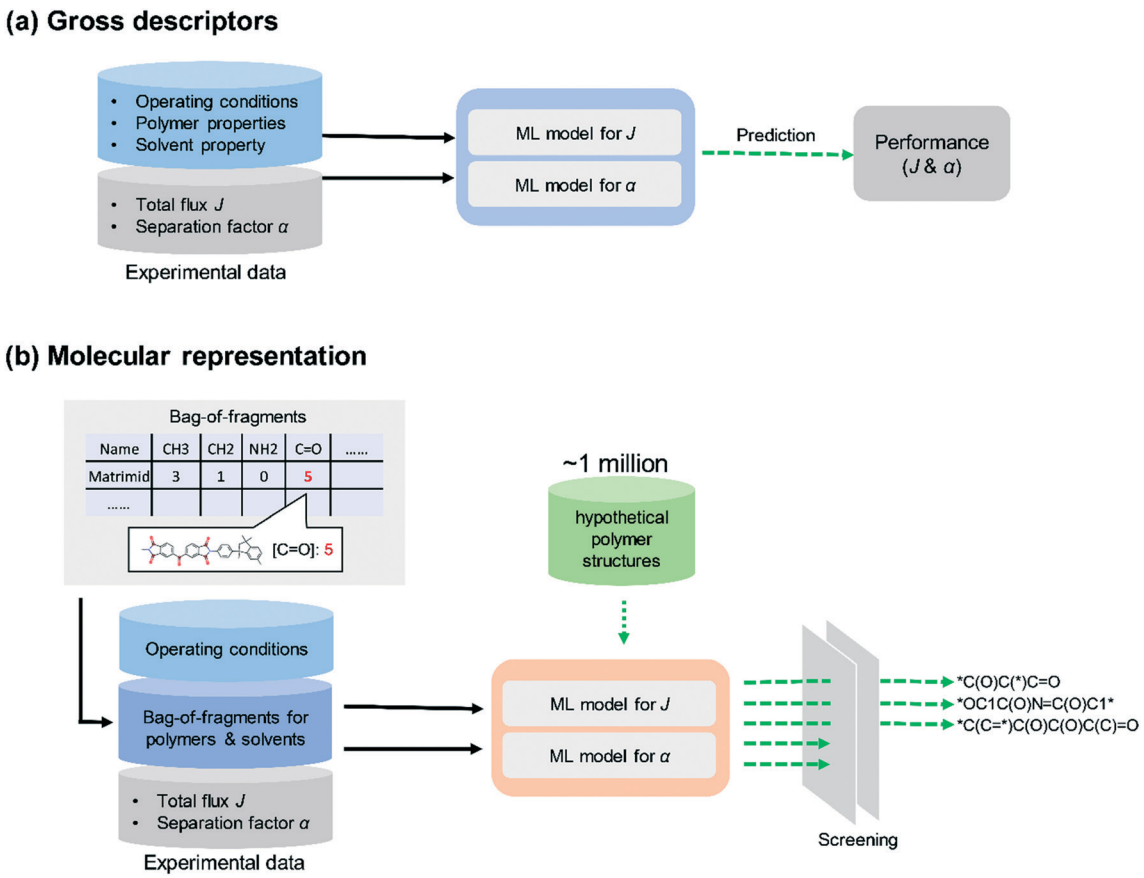


Fig. 3 ML for PV separation.[60] (a) A model based on gross descriptors (operating conditions, membrane and solvent properties) to predict total flux and separation factor. (b) A model based on molecular representation (bag-of-fragments for polymer and solvent) to predict and high-throughput screen ~1 million hypothetical polymers for PV separation of water/ethanol mixtures. Reproduced with permission: Copyright 2022, American Chemical Society.

ethanol separation. Using both a PV separation index >700 and a synthetic complex score as selection criteria, 10 polymers with a large portion of hydrophilic functional groups were identified as promising membrane materials for dehydration of water/ethanol mixtures.[60]

While PV has been demonstrated as a promising separation technology compared to conventional methods, its industrial adoption has been slow. This is in part due to several challenges in experimental PV, where ML studies are invaluable to derive more complete fundamental understanding of membrane performance. For example, MMMs generally have better PV performance than pristine polymer membranes, as the former leverage on the strengths of both polymers and fillers. However, the design and investigation of MMMs are not straightforward. Depending on the compatibility between polymers and fillers, a trade-off or enhancement in the separation factor and flux may be observed.[55] To comprehensively elucidate the effect of fillers, future ML studies should examine different combinations of polymers and fillers on permeant transport through various MMMs. It is also worthwhile to note that most current PV studies focus on water–organic mixtures, as highly selective separation is achievable owing to the large polarity difference between water and organic molecules. On the other hand, organic–organic separation is inherently challenging due to similar molecular properties. Therefore, the modeling exploration and rational design of novel membranes for such complex mixtures will be necessary. At the same time, membrane swelling and stability in organic media should also be quantitatively evaluated. As discussed above, existing ML studies for PV are largely limited to process level and preparation conditions for a few specified membranes. More efforts should also be directed to the curation of membrane structure databases with different membrane types (*e.g.*, polymer membranes and MMMs) and preparation conditions. Ideally, if these experimental challenges in PV are addressed, the resulting databases would facilitate holistic understanding of membrane structure–performance relationships, thus accelerating the identification and design of new membranes with high PV performance for water–organic and organic–organic mixtures.

5. Solvent recovery

In the chemical and pharmaceutical industries, large amounts of organic solvents are used for synthesis, cleaning

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and purification.[61] To produce active pharmaceutical ingredients, multiple reactions are typically performed in different solvents.[62] The total solvents used may account for 85% of raw materials by weight.[63] Their environmental factor, defined as the mass ratio of waste to desired product, stands at an alarming value of 25–100, generating considerably more chemical waste than the production of bulk chemicals and oil refining.[64,65] Consequently, solvent management has been recognized as one of the top key green engineering research areas for sustainable manufacturing.[66] Recently, organic solvent nanofiltration (OSN) has emerged as a robust technology for solvent recovery.[67] It also presents a novel paradigm in membrane processing, where selective molecular separation is achieved in organic media.[67] Extensive experimental studies have been conducted on OSN using different types of polymer membranes[68] and thin film (nano)composites.[69] Unlike the above discussed ML studies for water treatment and PV, ML for OSN-related properties has only been reported in the past 2–3 years, with primary focus on solubility parameters, polymer swelling and membrane performance.

5.1. Solubility parameters

Solubility parameters are of utmost importance in various chemical processes including membrane science.[70] The accurate quantitative predictions of these parameters are essential in assessing membrane swelling and separation performance in liquids. To predict the Hansen solubility parameters of polymers and solvents involved in OSN, Aspuru-Guzik and coworkers reported interpretable and versatile probabilistic ML models by leveraging on a variety of experimental results (Hansen solubility parameters) and simulation input data (COSMOtherm simulations and quantum chemistry calculations). Their ML models incorporated molecular descriptors such as electrostatic forces, σ-profiles, and molecular shapes and sizes, which demonstrated flexible predictions with measurable uncertainty bounds. Accurate predictions of solubility parameters were achieved, which outperformed well-established ML models.[71] After the curation of the Polymer Genome database containing experimental and computational data, Kim *et al.* constructed a hierarchical fingerprinting scheme from atomistic to morphological structural features towards quantitative description of polymers. They first utilized the recursive feature elimination algorithm to sequentially obtain an optimal set of features that intuitively affect polymer properties. Then, the Gaussian process regression (GPR) was employed to learn a nonlinear mapping between fingerprints and properties. Finally, their ML models were incorporated in a user-friendly online informatics platform to predict polymer properties such as the solubility parameter, glass transition temperature, bandgap, dielectric constant and atomization energy.[24] Given that well-established Hildebrand and Hansen solubility parameter models are widely used to estimate polymer solubilities in solvents, Venkatram *et al.* critically assessed the performance of these two models by comparing model predictions against actual experimental data. Using a benchmark set of 75 polymers, they concluded that the two criteria-based models provided similar and limited predictive accuracies for solvents and nonsolvents. For the Hildebrand model, nonsolvent predictions were highly accurate for all polymer classes, while solvent predictions were better for nonpolar polymers (70–75% accuracy) than for polar polymers (57% accuracy). This was attributed to the inherent description of nonpolar polymers by van der Waals forces in the Hildebrand model. For polar polymers, however, the polar and hydrogen-bonding interactions were not explicitly captured by the model.[72]

5.2. Polymer swelling

As shown in Fig. 4, ML for polymer swelling in liquids has been demonstrated through both classification and regression models. Different from the current phenomenological models based on solubility parameters, Ramprasad and coworkers proposed a data-driven paradigm for solvent selection in polymers. From a massive dataset of over 4500 polymers and their associated solvents/nonsolvents, they trained a deep neural network binary classifier with polymer (hierarchical structural features) and solvent (one-hot encoding) descriptors as inputs to qualitatively predict polymer swelling. By mapping the high-dimensional features to chemically meaningful latent space *via* principal component analysis (PCA), the spontaneous clustering of different polymer regions (soluble and insoluble) and solvent classes (nonpolar, aprotic and protic polar) was clearly observed. Compared to GPR using the Hildebrand parameters, their classification model demonstrated highly accurate determination of solvents/nonsolvents for new polymers.[73] Using the swelling of OSN and PDMS membranes as illustrations, Xu and Jiang constructed regression models to quantitatively estimate polymer swelling in liquids. Through kernel ridge regression, an ML model based on chemically intuitive descriptors such as solubility parameters and solvent properties was developed to predict polymer swelling. A multimodal swelling behavior was observed for OSN membranes, while the solubility parameter and solvent geometry were identified as key properties for PDMS swelling. Then, a molecular representation *via* a sum-of-fragment approach was proposed to learn the chemical space of polymer swelling. Through data augmentation and statistical analysis, the ML model based on molecular representation also demonstrated accurate predictions and provided useful insights into polymer swelling.[74]

5.3. Membrane performance

The complex interactions occurring during OSN pose challenges in accurate prediction of membrane performance. Very recently, a few ML studies on OSN membrane

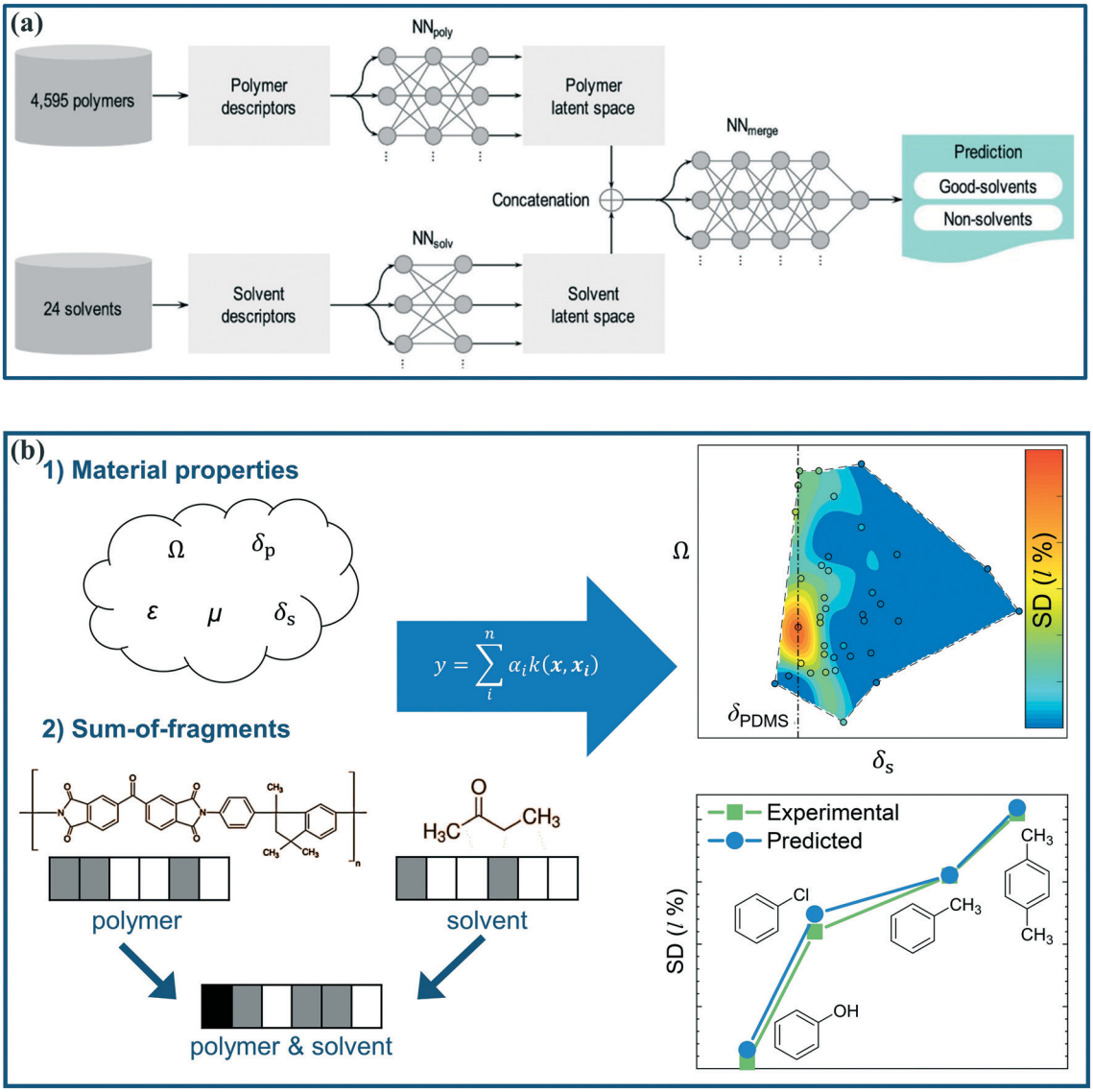


Fig. 4 ML for polymer swelling. (a) Classification model for solvents and nonsolvents through neural networks.[73] (b) Regression models through kernel ridge regression using material properties and molecular representation.[74] Reproduced with permission: Copyright 2020, American Chemical Society.

performance have been reported. As an alternative to existing phenomenological models, Skiborowski and coworkers used ML to identify membrane-specific models for predicting solvent permeance.[75] Particularly, they developed a hybrid stochastic–deterministic approach to obtain accurate solvent permeance models. Using various polymer and solvent properties (*e.g.*, size and polarity), various predictive models were formulated by genetic programming, while the nonlinear parameter regression was achieved through global deterministic optimization algorithms. They applied their data-driven approach to three case studies, where pure and mixed solvent permeances through PDMS and ceramic membranes were predicted with superior accuracy and precision over existing transport models.[75] With automatic development of membrane-specific models, they extended to predict solute rejections in pure and mixed solvent systems. By considering an additional 29 different solvent and solute descriptors, both solute-independent and solute-specific rejection models were developed for PuraMem S600 and 280 membranes. In the respective six pure polar and nonpolar solvents, varying complexity in solute rejection models was observed for different solvent classes. By leveraging on solute rejection data from pure solvent and equimolar binary solvent mixtures, their approach rapidly predicted membrane rejection maps for other mixed solvent systems, which were consistent with experiments.[76]

Szekely and coworkers also employed ML to predict OSN membrane performance in terms of permeance and rejection, as shown in Fig. 5, by curating and standardizing 38 430 OSN data points with more than 18 parameters. In the data acquisition step, they extracted the metadata from the available literature and transformed them into numeric or

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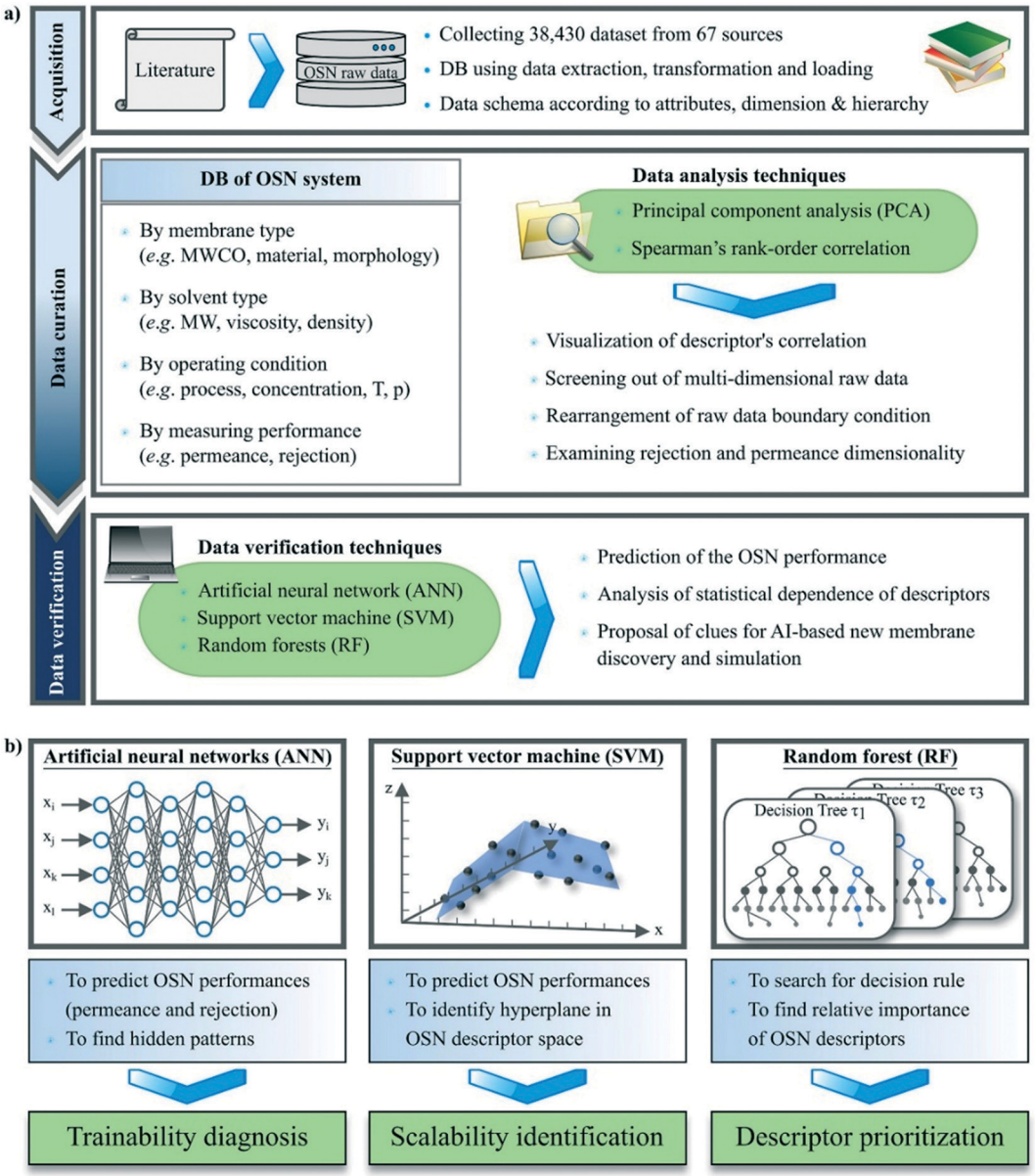


Fig. 5 (a) Flowchart to predict OSN membrane performance with (b) three different ML algorithms.[77] Reproduced with permission: Copyright 2020, Elsevier.

symbolic values to organize OSN raw data according to various attributes, dimensions and hierarchies. The OSN database consisted of membrane and solvent types, operating conditions and performance data, which were analyzed critically by Spearman's rank-order correlation and PCA. Through such exploratory data analysis, the multi-dimensional database was visualized and examined to reveal the effects and correlations of different descriptors on membrane performance. They further performed detailed PCA to reduce data dimensions and identify important parameters affecting membrane performance, namely the membrane molecular weight cut-off (MWCO), solute molecular weight and concentration, solvent factor, temperature and process configuration. Out of these parameters, the solvent factor and solute molecular weight and concentration were three parameters with top relative importance. Thereafter, they employed three different ML algorithms (*i.e.*, ANN, SVM and random forest) to analyze the statistical dependence of descriptors and predict OSN performance of new membranes. Overall, their ML models demonstrated excellent trainability diagnosis, scalability identification and descriptor prioritization for OSN membranes; decision rules and hidden patterns in the descriptor space were elucidated, while superior predictive accuracy as high as 98% for permeance and 91% for rejection was achieved.[77] Moreover, they employed robust medium-throughput experiments for three commercial DuraMem polyimide membranes to measure solute rejection in OSN.

From two different operating pressures in methanol, the generated rejection dataset of 336 selected solutes was correlated with 26 molecular descriptors. A novel rejection prediction scheme based on the Tanimoto similarity of the solute structure was also proposed and made publicly available online on the OSN Database website.[78] By leveraging on the OSN Database, they further developed two ML models to predict solute rejection directly from molecular structures. Based on quantitative structure–activity relationships, partial least squares regression was first combined with a GA to identify molecular descriptors (*i.e.*, charge, electronic and topological properties) critical to solute rejection. In addition, a message-passing neural network was also constructed to visualize and estimate the contributions of various functional groups in solutes to their rejection in methanol through three commercial polyimides.[79]

To date, ML studies of polymer membranes for non-aqueous separation (*e.g.*, PV and OSN) are limited due to many existing challenges. Given their large molecular sizes and high viscosities, organic solvents typically have slow permeation. Moreover, polymer membranes in non-aqueous media exhibit more complex structural dynamics due to swelling. To realistically describe these phenomena, advanced ML methods should be developed to thoroughly characterize membrane stability in a non-aqueous phase in terms of chemical, mechanical and thermal properties. At present, standardized experimental design and protocols that reliably compare membranes under industrially relevant operating conditions are not well established. It has been recommended that detailed descriptions of membrane fabrication, characterization and performance measurement should be provided.[80] On this basis, the membrane community can not only develop membrane processing–structure–performance relationships, but also motivate more accurate and physically meaningful ML. For future ML exploration, predictive models based on various molecular representations of membranes and solvents are more attractive and should be developed.[81] Together with improved statistical analysis, useful insights from these molecular structure-based ML models would accelerate the rational design of new polymer membranes with high-performance separation for non-aqueous mixtures.

6. Future directions

This review summarizes recent ML studies of polymer membranes for liquid separation. When adopted, this computational approach can successfully unravel fundamental insights, predict separation performance and provide membrane design principles. While a number of ML studies have been reported to optimize process conditions for water treatment, such efforts at the membrane level are limited. For PV, ML studies are scarce at both process and membrane levels. As a burgeoning process, OSN has only recently witnessed the applications of ML to predict solubility parameters, polymer swelling and membrane performance.

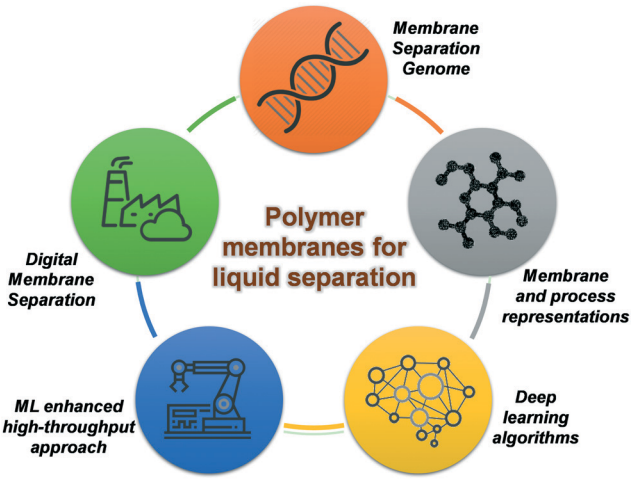


Fig. 6 Suggested directions for future ML exploration.

The complete description of practical liquid separation through polymer membranes is not straightforward as the process covers many events and phenomena occurring at varying length and time scales. Particularly, the separation performance is highly dependent on a wide variety of intricately connected membrane properties and processes: chemical structures of monomers, synthesis of polymers, fabrication methods for membranes with(out) fillers, complex interplay of interactions among membrane–solvent–solute, and operating conditions of the membrane process, just to name a few. Currently, the holistic consideration of all these factors in a given study remains technologically challenging, time-consuming and resource intensive. To overcome these challenges, several directions are suggested as shown in Fig. 6 for future ML exploration of polymer membranes for liquid separation.

(1) Membrane Separation Genome

First and foremost, databases are the most important resources in ML studies. As previously discussed, the complete reporting of membrane properties and performance remains a technological challenge due to the large diversity of polymer types, experimental fabrication methods and separation conditions of polymer membranes. Furthermore, the current reporting of fabrication methods, characterization techniques, additives and (non)solvents has been inconsistent, and the resulting data are mostly scattered in the literature. While there exist well-established handbooks and online polymer databases, they are only limited to traditional thermophysical properties such as glass transition temperatures and specific volumes. The scarcity and inconsistency of membrane data could hamper future ML efforts for membrane separation. Therefore, it is essential to construct databases that comprehensively include experimental and even computational properties of

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membranes, which we call the *Membrane Separation Genome*. The curation of standardized and extensive experimental databases is recommended, where membrane types, fabrication methods and separation conditions are systematically compiled. Firstly, membrane synthesis methods (*e.g.*, interfacial polymerization, casting and phase inversion) with associated reagents and conditions (*e.g.*, monomers, additives, temperature, and agitation) should be reported for different membrane structures (*e.g.*, chemical structures of monomers and linkers). The membrane characterization methods (*e.g.*, microscopic and spectroscopic techniques) and membrane structural and physicochemical properties (*e.g.*, membrane thickness, free volume, contact angle and swelling degree in liquids) need to be included in the databases. Subsequently, the experimental procedure for liquid separation should be explicitly provided in terms of operating conditions such as the temperature, transmembrane pressure, solvent type, solute type and concentration. In addition, the separation performance metrics (*e.g.*, permeability, rejection and MWCO) are also required. To investigate the long-term membrane phenomena (*e.g.*, aging, plasticization and fouling) in practical liquid separation, time-dependent membrane structural and physicochemical properties (*e.g.*, change in free volume with time) and separation performance (*e.g.*, change in solvent permeability and solute selectivity with time) should also be included in the databases. For the liquid separation discussed above, the main differences lie in the operating conditions (*e.g.*, temperature and pressure in the feed and permeate side) and the nature and composition of the feed (*e.g.*, water, organic solvent and solute). Such detailed information should also be clearly stated in the databases for future ML studies for the respective separation. Not only should these databases collect the important properties of existing polymer membranes, but they should also be readily available and allow for the incorporation of new membranes and their associated fabrication and operating conditions. On the other hand, computational databases of amorphous polymer membranes are currently very limited, in contrast to well-established ones for crystalline materials. Such databases should include molecular models of polymer membranes and equilibration schemes involved in model constructions. These resources will be invaluable to the computational reproducibility and investigation of polymer membranes for liquid separation. Through these computational databases, detailed ML and high-throughput screening can be performed to identify promising polymer membranes.

(2) Membrane and process representations

After the curation of membrane databases, the subsequent step involves the transformation of inputs into digitized formats suitable for ML. Particularly, various representations from membrane to process levels should be appropriately developed and utilized for ML. At the membrane structural level, the amorphous nature of polymer membranes needs to be first characterized. While common gross membrane properties (*e.g.*, glass transition temperature and solubility parameter) relevant to membrane separation are desirable descriptors, they might not be easily accessible and measured experimentally. Alternatively, molecular representation is more general in capturing the chemical structures of polymers and liquids. To numerically represent them, their chemical structures should be digitally encoded *via* various molecular representation approaches such as SMILES and molecular graphs.[81] Further novel development of ML methods and descriptors would be required to quantify the unique structural architectures of various state-of-the-art polymer membranes (*e.g.*, block polymer membranes,[82] thin film composites[83] and MMMs[84,85]) for liquid separation. Indeed, recently proposed numerical representations of polymers for a range of applications may also be relevant to the description of polymer membranes. For example, the single-chain sequence of a coarse-grained polymer was featurized into a convolved image using convolution filters to predict the radius of gyration.[86] Using context-free-grammar parse trees and stochastic lazy attributes, the syntax of the SMILES language and the semantics of polymers were incorporated in variational encoders to inverse design polymers with high thermal and electrical stability.[87] Copolymer fingerprints consisting of SMILES of polymer repeat units and composition information were employed as inputs for multitask neural networks to predict the glass transition, melting and degradation temperatures of homopolymers and copolymers.[88] Under the context of known polymer sequence or composition information, further featurization strategies for copolymer systems also include one-hot encoding of constitutional units, molecular fingerprints and descriptor vectors.[89] The monomer structure of polymers can be modeled by molecular graphs comprising sets of nodes (atoms) and edges (chemical bonds) before the training of the graph convolutional network.[90]

At the membrane fabrication level, synthesis procedures are typically reported in a text form of the experimental section that describes fabrication methods, characterization techniques and additives involved. The digitalization of the text into a computer-readable form will be necessary for ML. Apart from temperature and pressure, membrane processing should also be quantitatively described in terms of the operating mode (*e.g.*, cross-flow and dead-end), membrane modules used (*e.g.*, hollow fiber and spiral wound), *etc.* The numerical featurization of membrane structural information, fabrication and processing conditions will be essential but not straightforward. Depending on the target performance, one may begin with a list of readily available and uncorrelated features with relevant information. During ML model construction, important input parameters critical to the target performance may be identified and even motivate the development of novel representations. Overall, holistic representations should not only be physically and chemically interpretable, but also allow accurate ML predictions and

unravel fundamental insights into membrane fabrication and processing.

(3) Deep learning algorithms

After the curation of membrane databases and the featurization of inputs, robust ML algorithms are necessary to establish quantitative relationships that map these inputs to separation performance. Currently, various state-of-the-art ML algorithms are widely available, particularly the ANN has emerged as a versatile and accurate algorithm capable of deep learning.[21] As previously mentioned, while the ANN has been employed for membrane separation, it is limited mostly to aqueous separation and at the process level. Further efforts should also be directed to explore a myriad of neural network architectures for the ML of various membrane properties and separation performance. For example, the recurrent neural network (RNN) is well-suited to handle sequential inputs (*e.g.*, text and time-series data) to predict the next scenario. Membrane synthesis procedures contain rich and complex text data, which may be modeled by the RNN to make prediction for the subsequent membrane fabrication. Based on previous time-series data of a process plant, the RNN may also provide temporal prediction for the forecasting and process control of membrane separation. On the other hand, the structural characterization of membranes can be achieved with the convolutional neural network (CNN). For instance, the CNN is useful in the description of membrane morphologies such as the pore size distribution and spatial distribution, and may be used for performance prediction based on membrane structural architectures. Another class of CNNs is the graph CNN that has proven successful in the modeling of atoms and bonds of molecules in terms of nodes and edges.[91] Therefore, the chemical structures of polymer membranes and liquids can be readily transformed into molecular graphs for ML. Furthermore, by coupling neural networks and latent space modeling, deep generative models (*e.g.*, variational autoencoders and generative adversarial networks) have demonstrated tremendous success in recent years for many materials.[81] These deep learning algorithms may also hold enormous promise for membrane separation in terms of novel generation and inverse design of optimized membrane structures and fabrication and operating conditions. While neural networks have been discussed here to address the aforementioned membrane issues, further algorithmic development in other classes of ML is also desired. On the other hand, it is worthwhile to highlight existing challenges to be addressed before ML becomes more prevalence in polymer membranes for separation. Firstly, future researchers are encouraged to publicly report database, code and algorithm resources as much as possible. Secondly, to increase data volume for deep learning, transfer learning may be implemented using pre-trained models while data augmentation may also be explored by generating new synthetic data from existing data. Lastly, the trained models should be further optimized to obtain ideal operating conditions and maximal separation performance, followed by the validation of ML prediction by additional experiments.

(4) ML enhanced high-throughput approach

The advent of automated instruments has promoted the widespread demonstration of high-throughput experiments in the fields of chemistry and process development.[92,93] Meanwhile, ML can leverage on the large databases generated from these experiments. Indeed, the applications of ML in high-throughput experimental data have accelerated the synthesis and development of new materials and catalysts.[94] Likewise, membrane development for separation can also benefit from the ML enhanced high-throughput approach. To date, however, high-throughput screening has been largely underexploited in membrane research.[95] At the membrane level, there were a few high-throughput studies including the investigation of OSN membranes,[96] phase inversion parameters of polyimides,[97] process conditions of hollow-fiber membranes,[98] and swelling of copolymer membranes for pervaporation.[99] At the process level, the ML enhanced high-throughput approach was reported to optimize various process units for water treatment plants.[44–46] Nevertheless, these high-throughput studies are typically limited to a few specified classes of polymer membranes.

To leverage on the ML enhanced high-throughput strategy, first, comprehensive and systematic synthesis and fabrication of polymer membranes that cover large chemical diversity and various fabrication conditions are highly desirable. For example, combinatorial synthesis of membranes with components of varying compositions may be performed *via* high-throughput approaches such as vial-based methods, well plates and robotic platforms. Next, physically meaningful ML models can be developed to explore the diverse design space spanning from membrane materials to fabrication and operating conditions for high-performance separation. In this regard, Bayesian optimization is a robust ML method that is able to learn from past experiments to inform the next optimized synthesis of various materials.[100] By employing the Gaussian process surrogate model, this learning method may be applied to direct the selection of membrane materials and fabrication and operating conditions towards optimized separation performance. In addition, reinforcement learning, where decisions are made based on actions and rewards in a prescribed environment,[101] may be adopted for membrane synthesis planning to make decisions on fabrication and operating conditions towards the optimization of membrane materials. Ultimately, the ML enhanced high-throughput approach can not only identify the best synthesis planning to accelerate membrane development, but also reduce the high cost involved in trial-and-error experiments.

(5) Digital Membrane Separation

We envision *Digital Membrane Separation* as part of the Smart Factory of the Future to realize the ongoing industrial

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revolution, Industry 4.0. In Industry 4.0, the Smart Factory promises high efficiency in the allocation and utilization of limited resources while rapidly adapting to the ever-changing demands of the manufacturing industry. The key enabler of such advanced manufacturing is the application of information technology, which comprises cyber-physical systems (CPS), cloud computing and Internet of Things (IoT).[102] Firstly, the CPS seamlessly integrates different physical processes to a central computing system.[103] In this interconnected network, the embedded computers perform real-time monitoring and computation to coordinate different physical machines that provide feedback loops for immediate rectification. Next, cloud computing encompasses the secured databases and high-performance computations at a remote platform through the internet.[104] Such computational capability facilitates easy accessibility and sharing of data and information along a value chain, thus reducing the associated operational costs. In addition, the IoT also allows intelligent communications among self-driven machines that are dynamically connected in a global network infrastructure.[105] Given the rapid development in tracking devices, the manufacturing process can be constantly monitored and predicted for its future performance. Therefore, to realize the *Digital Membrane Separation* in the Smart Factory, a multiscale computational approach to drive separation from the membrane and process levels to the industrial level should be adopted. By leveraging on the previously discussed approaches, these data-driven strategies should form a ML framework for the *Digital Membrane Separation*. For example, through cloud computing, large databases of polymer structures, fabrication and operating conditions and separation performance are continuously updated, analyzed and accessed online. These comprehensive databases will be invaluable resources to address various existing membrane challenges such as the elucidation of complex solute–solvent–membrane interactions and the synthesis of advanced membrane materials with high permeability and selectivity. Autonomous mobile robots should be intensively involved to perform high-throughput membrane synthesis, characterize structural integrity of polymer membranes, and monitor real-time membrane separation conditions and performance, just to name a few. By synergizing these automated robotic capabilities with optimization methods, self-optimizing membrane units can be developed to dynamically meet different separation objectives. These intelligent machines can also communicate among each other and provide instantaneous feedback to the central computing system for the optimization of membrane fabrication and separation performance. Potentially, the digitalization of the membrane process may drive autonomous separation with enhanced connectivity from the membrane to process levels, higher operational efficiency, and more predictable and optimized separation performance.

The above suggested directions provide new opportunities and challenges to holistically apply ML for membrane separation (indeed applicable to both liquid and gas separation). Fundamental guidelines can be offered for the rational design of materials at the membrane level. Meanwhile, ML at the process level can facilitate membrane module design and plantwide process optimization. Ultimately, ML can bridge the technological gap between the membrane and process levels, thereby enabling more viable development and commercialization of new polymer membranes with high separation performance.

Conflicts of interest

There are no conflicts to declare.

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